

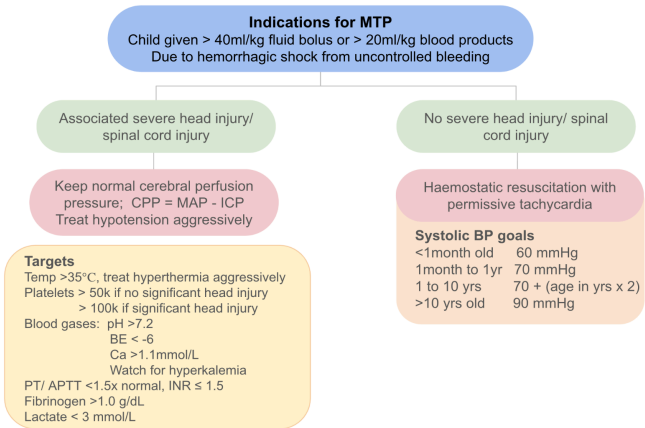
Paediatric Massive Transfusion Protocol (MTP)

Massive transfusion in children is defined as transfusion of >40 ml/kg (blood volume in children >1 month old is approximately 80 ml/kg)

MTP hotline blood bank: dial extension _____ (check with your own institution)

See below for eg of a MTP activation chart.

Paediatric Massive Transfusion Protocol



Indications for MTP

Child given >40ml/kg fluid bolus or >20ml/kg blood.
Primary cause of shock is haemorrhagic shock from uncontrolled bleeding.

Associated severe head injury/ spinal cord injury

Maintain normal cerebral perfusion pressure
Cerebral perfusion pressure
= MAP - ICP

NO associated severe head injury/ spinal cord injury

Haemostatic resuscitation with permissive tachycardia

Systolic blood pressure goals:

<1month: SBP = 60mmHg

1month – 1year: SBP = 70mmHg

1 to 10years old: SBP = $70 + ((\text{age in years}) \times 2)$

>10 years old: SBP = 90mmHg

Monitor for:

1. Temperature >35 and treat hyperthermia aggressively
2. Treat hypotension aggressively in children with head injury or spinal cord injury
3. Platelets
 - Keep platelets $\geq 50 \times 10^9/L$ if NO significant head injury present
 - Keep platelets $\geq 100 \times 10^9/L$ in presence of significant head injury
4. Blood gases/ electrolytes
 - pH >7.2
 - Base excess < -6
 - Ca $^{2+} > 1.1\text{mmol/L}$
 - Monitor for hyperkalemia
5. Coagulation profile
 - PT/ aPTT $< 1.5\times$ normal
 - INR ≥ 1.5
6. Fibrinogen $>1\text{g/L}$
7. Lactate $< 3 \text{ mmol/L}$